

Concrete Next Steps

This note records the most actionable next moves after compiling the current Faber–Krahn stability notes.

1 Prove the Finite-Perimeter Level-Set Identity

This is the cleanest low-risk theorem to write next.

Target statement: for every open set Ω of finite measure, with torsion function $u = u_\Omega$, the identity

$$\frac{2}{|\Omega|}(E(\Omega) - E(B)) = \frac{1}{|\Omega|} \int_0^{\|u\|^\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt$$

holds with $E_t = \{u > t\}$, $m(t) = |E_t|$, and $P(t) = P(E_t)$ for a.e. t .

The proof should be organized as follows:

1. Use coarea for Sobolev functions to get finite perimeter of E_t for a.e. t and

$$-m'(t) = \int_{\partial^* E_t} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}.$$

2. Prove the weak divergence identity

$$\int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1} = m(t)$$

by testing $-\Delta u = 1$ with Lipschitz truncations of $(u - t)_+$.

3. Define $I(s) = \int_{\{u > u^*(s)\}} u$ and justify

$$I'(s) = u^*(s), \quad I''(s) = -1/a(u^*(s))$$

first where m is differentiable, then in the distributional sense.

4. Finish by the convexity-gap identity

$$G'(L) - G(L)/L = L^{-1} \int_0^L s G''(s) ds.$$

This gives a polished standalone theorem, independent of the selection principle.

2 Import the STFT Stability Paper's Profile-Gap Method

The new file `s00222-024-01248-2-3.pdf` suggests a sharper one-dimensional starting point. In that paper, stability is driven by the area between the rearranged profile $u^*(s)$ and the optimizer profile e^{-s} .

For torsion, define

$$v(s) := u_\Omega^*(s), \quad b(s) := u_B^*(s),$$

where B is the ball with $|B| = |\Omega| = L$. Since

$$b'(s) = -\frac{1}{n^2 \omega_n^{2/n} s^{1-2/n}}$$

and the level-set inequality gives $v'(s) \geq b'(s)$, the profile gap

$$h(s) := b(s) - v(s)$$

is nonnegative and nonincreasing. Moreover

$$\int_0^L h(s) ds = 2(E(\Omega) - E(B)).$$

Thus

$$0 \leq b(s) - v(s) \leq \frac{2(E(\Omega) - E(B))}{s}.$$

This gives an explicit way to choose near-boundary levels: for $s = L - \eta$, if $E(\Omega) - E(B) \ll \eta$, then

$$t_\eta := v(L - \eta)$$

is comparable to the ball's boundary-layer height and

$$|\Omega \setminus \{u > t_\eta\}| = \eta.$$

The next lemma to prove is then a near-boundary good-level lemma: among $s \in [L - 2\eta, L - \eta]$, find one level with controlled $D_H + D_I$. For selected minimizers, free-boundary regularity should convert the corresponding height control into geometric boundary-layer control.

3 Quantify Graph Entry for Selected Minimizers

The most direct continuation of the BDV route is the following theorem.

Let $U \subset B_R$ be a selected minimizer of the penalized functional, with the BDV density estimates and Lipschitz Bernoulli coefficient q_U . Then there exists an explicit threshold $\varepsilon_0(N, R, \tau)$ such that

$$\alpha(U) \leq \varepsilon_0(N, R, \tau) \implies \partial U \text{ is a global } C^{1,\gamma} \text{ graph over } \partial B_1.$$

The proof should follow this deterministic chain:

1. From density plus BDV Lemma 4.2,

$$d_H(\partial U, \partial B_1) \leq C\alpha(U)^{1/(2N)}.$$

2. Fix $\mu < \bar{\mu}$, where $\bar{\mu}$ is the Alt–Caffarelli flatness threshold, and choose the corresponding scale $\rho(\mu)$.

3. Impose

$$C\alpha(U)^{1/(2N)} \leq \mu\rho(\mu).$$

Then every ball $B_{4\rho(\mu)}(x_0)$ centered near ∂B_1 sees a sufficiently flat free boundary.

4. Apply Alt–Caffarelli locally and patch the finite cover of ∂B_1 into a global graph.

This would turn the compactness phrase “for j large” into a concrete smallness condition.

4 Use a Hybrid Route: Apply Level Sets to Selected Minimizers

The level-set route has a boundary-layer obstruction for arbitrary domains. The selected minimizers are much better objects: their free boundaries have nondegeneracy, density estimates, and a Bernoulli condition with controlled q_U .

This suggests a hybrid theorem:

For a selected minimizer $U = \{u > 0\}$, the Saint-Venant deficit controls the isoperimetric deficit of near-boundary level sets $\{u > t\}$, uniformly for $0 < t < t_0(N, R, \tau)$.

The expected mechanism is:

1. Since q_U is bounded above and below on ∂U , the coarea factor $a(t) = \int_{\{u=t\}} 1/|\nabla u|$ is uniformly controlled for small t .
2. For small t , the level surface $\{u = t\}$ is a parallel perturbation of ∂U , so

$$|U \setminus \{u > t\}| \simeq t \int_{\partial U} \frac{1}{q_U}, \quad P(\{u > t\}) = P(U) + O(t).$$

3. The weighted level-set identity then gives a genuinely near-boundary good level, not merely an interior good level.
4. Letting $t \downarrow 0$ should recover a bound on the isoperimetric deficit of U itself.

If this works, it may shorten the route from “selected minimizer” to “near-ball” and reduce reliance on the full smooth-convergence compactness argument.

5 Boundary-Layer Asymmetry Lemma

A small but useful lemma to prove independently:

If $E_t = \{u > t\}$, $s = |E_t|$, $L = |\Omega|$, and $\eta = L - s$, then

$$\mathcal{A}(\Omega) \leq C \frac{\eta}{L} + C \mathcal{A}(E_t)$$

after the harmless volume normalization of the optimal ball for E_t .

Proof: take an optimal ball B_s for E_t , dilate it to a ball B_L with volume L , and estimate

$$|\Omega \Delta B_L| \leq |\Omega \setminus E_t| + |E_t \Delta B_s| + |B_s \Delta B_L|.$$

This isolates the exact missing ingredient in the pure level-set route: finding a good level with both η small and $\mathcal{A}(E_t)$ controlled.

6 Superlevel Replacement, Copied from the STFT Strategy

The STFT stability proof first replaces the arbitrary set Λ by the matched superlevel set

$$A_\Lambda = \{u_F > u_F^*(|\Lambda|)\},$$

proves stability of A_Λ , and then transfers stability back to Λ by a transport comparison.

For torsion, the analogous replacement is

$$\Omega \rightsquigarrow E_t = \{u_\Omega > t\}, \quad |E_t| = |\Omega| - \eta.$$

The transfer back to Ω is exactly the boundary-layer asymmetry lemma. The new work is to prove that the chosen E_t is geometrically good. The profile-gap lemma chooses t in the correct near-boundary range; the exact level-set deficit identity gives averaged control of $D_H + D_I$; selected minimizer regularity supplies the missing uniform estimates needed to convert averages into a useful level.

7 Serrin-Stability Input to Look For

The level-set deficit gives a weighted L^2 variance:

$$\int_{\Sigma_t} \frac{(|\nabla u| - \bar{f})^2}{|\nabla u|} = \frac{m(t)}{P(t)^2} D_H(t).$$

Most available Serrin-stability theorems use stronger boundary norms, such as oscillation or Lipschitz seminorms of u_ν . The bridge should be

$$\|f - \bar{f}\|_{L^\infty(\Sigma_t)} \leq C \|f - \bar{f}\|_{L^2(\Sigma_t)}^\theta [f]_{C^\gamma(\Sigma_t)}^{1-\theta},$$

valid on uniformly $C^{1,\gamma}$ level surfaces. Selected minimizers are the right setting for this interpolation because their free-boundary regularity provides the missing C^γ control.

Relevant Serrin-stability references to check first:

- Brandolini–Nitsch–Salani–Trombetti, *On the stability of the Serrin problem*, JDE 2008.
- Ciraolo–Magnanini–Vespri, *Holder stability for Serrin's overdetermined problem*, Ann. Mat. Pura Appl. 2016.
- Magnanini–Poggesi, integral-identity approaches to Serrin stability.

The likely outcome is that the existing theorems do not plug in directly, but their interpolation and integral-identity methods should be adaptable to the weighted L^2 defect coming from D_H .

8 Analytic/Native-Space Calculation

The speculative but concrete GGRT-inspired calculation is:

1. Work after BDV selection, so the domain is a smooth near-ball

$$\partial\Omega_\varepsilon = \{(1 + \varepsilon\varphi(\theta))\theta\}.$$

2. Replace torsion by the harmonic object

$$h_\varepsilon = u_{\Omega_\varepsilon} + \frac{|x|^2}{2N}.$$

For the ball this is constant, and the first variation is the harmonic extension of the boundary graph.

3. For fixed $s \in (0, |B|)$, compute the second variation of

$$\mathcal{K}_s(\Omega) = \int_{\{u_\Omega > u_\Omega^*(s)\}} u_\Omega.$$

4. Diagonalize in spherical harmonics. The $k = 0$ mode should vanish because of volume normalization, and $k = 1$ because of translation. The question is whether the coefficients for $k \geq 2$ control the $H^{1/2}$ -norm.

In dimension two, also test the holomorphic formulation

$$F_\Omega(z) = \partial_z u_\Omega(z) + \frac{\bar{z}}{4},$$

which is holomorphic in Ω and vanishes for centered disks. Pulling this object back to the disk may give a Bergman/Hardy-space version of the GGRT Fock-space computation.